Gauss-Jordan Method

The method is based on the idea of reducing the given system of equations **Ax = b,** to a diagonal system of equations **Ix = d,** where **I** is the identity matrix, using elementary row operations. We know that the solutions of both the systems are identical. This reduced system gives the solution vector **x**. This reduction is equivalent to finding the solution as $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

$$[\mathbf{A}|\mathbf{b}] \xrightarrow{\text{Gauss-Jordan method}} [\mathbf{I} \mid \mathbf{X}]$$

In this case, after the eliminations are completed, we obtain the augmented matrix for a 3 × 3 system as

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & d_1 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 1 & d_3 \end{array}\right] \qquad (1)$$

and the solution is $x_i = d_i,\ i = 1, 2, 3.$

Elimination procedure The first step is same as in Gauss elimination method, that is, we make the elements below the first pivot as zeros, using the elementary row transformations. From the second step onwards, we make the elements below and above the pivots as zeros using the elementary row transformations. Lastly, we divide each row by its pivot so that the final augmented matrix is of the form (1). Partial pivoting can also be used in the solution. We may also make the pivots as 1 before performing the elimination.

Let us illustrate the method.

Example *Solve the following system of equations*

$$x_1 + x_2 + x_3 = 1$$
$$4x_1 + 3x_2 - x_3 = 6$$
$$3x_1 + 5x_2 + 3x_3 = 4$$

using the Gauss-Jordan method (i) without partial pivoting, (ii) with partial pivoting.

Solution We have the augmented matrix as

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 4 & 3 & -1 & 6 \\ 3 & 5 & 3 & 4 \end{array}\right]$$

We perform the following elementary row transformations and do the eliminations.

$$R_2 - 4R_1, R_3 - 3R_1 : \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & -5 & 2 \\ 0 & 2 & 0 & 1 \end{array}\right].$$

$$R_1 + R_2, R_3 + 2R_2 : \left[\begin{array}{ccc|c} 1 & 0 & -4 & 3 \\ 0 & -1 & -5 & 2 \\ 0 & 0 & -10 & 5 \end{array}\right].$$

$$R_1 - (4/10)R_3, R_2 - (5/10)\, R_3 \; : \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & -1 & 0 & -1/2 \\ 0 & 0 & -10 & 5 \end{array}\right].$$

Now, making the pivots as 1, $((-R_2), (R_3/(-10)))$ we get

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & -1/2 \end{array}\right].$$

Therefore, the solution of the system is $x_1 = 1$, $x_2 = 1/2$, $x_3 = -1/2$.

Example Solve the following system by elimination method.

$$x + 3y = 7$$
$$3x + 4y = 11$$

Solution

We multiply the first equation by -3, and add it to the second equation.

$$-3x - 9y = -21$$
$$3x + 4y = 11$$
$$\overline{\qquad\qquad\qquad}$$
$$-5y = -10$$

By doing this we transformed our original system into an equivalent system:

$$x + 3y = 7$$
$$-5y = -10$$

We divide the second equation by -5, and we get the next equivalent system.

$$x + 3y = 7$$
$$y = 2$$

Now we multiply the second equation by -3 and add to the first, we get

$$x = 1$$
$$y = 2$$

Example Solve the following system by the Gauss-Jordan method.

$$x + 3y = 7$$

$$3x + 4y = 11$$

Solution

The augmented matrix for the system is as follows.

$$\left[\begin{array}{cc|c} 1 & 3 & 7 \\ 3 & 4 & 11 \end{array}\right] \quad \begin{bmatrix} x + 3y = 7 \\ 3x + 4y = 11 \end{bmatrix}$$

We multiply the first row by -3, and add to the second row.

$$\left[\begin{array}{cc|c} 1 & 3 & 7 \\ 0 & -5 & 10 \end{array}\right] \quad \begin{bmatrix} x + 3y = 7 \\ -5y = 10 \end{bmatrix}$$

We divide the second row by -5, we get,

$$\left[\begin{array}{cc|c} 1 & 3 & 7 \\ 0 & 1 & 2 \end{array}\right] \quad \begin{bmatrix} x + 3y = 7 \\ y = 2 \end{bmatrix}$$

Finally, we multiply the second row by -3 and add to the first row, and we get,

$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right] \quad \begin{bmatrix} x = 1 \\ y = 2 \end{bmatrix}$$

Example Solve the following system by using the Gauss-Jordan elimination method.

$$\begin{cases} x + y + z = 5 \\ 2x + 3y + 5z = 8 \\ 4x + 5z = 2 \end{cases}$$

Solution: The augmented matrix of the system is the following.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 2 & 3 & 5 & 8 \\ 4 & 0 & 5 & 2 \end{array}\right]$$

We will now perform row operations until we obtain a matrix in reduced row echelon form.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 2 & 3 & 5 & 8 \\ 4 & 0 & 5 & 2 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 3 & -2 \\ 4 & 0 & 5 & 2 \end{array}\right]$$

$$\xrightarrow{R_3 - 4R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 3 & -2 \\ 0 & -4 & 1 & -18 \end{array}\right]$$

$$\xrightarrow{R_3+4R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 13 & -26 \end{array}\right]$$

$$\xrightarrow{\frac{1}{13}R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

$$\xrightarrow{R_2-3R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

$$\xrightarrow{R_1-R_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 7 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

$$\xrightarrow{R_1-R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

From this final matrix, we can read the solution of the system. It is

$$\boxed{x = 3, \quad y = 4, \quad z = -2.}$$

Example Solve the following system by using the Gauss-Jordan elimination method.

$$\begin{cases} A + B + 2C = 1 \\ 2A - B + D = -2 \\ A - B - C - 2D = 4 \\ 2A - B + 2C - D = 0 \end{cases}$$

Solution: We will perform row operations on the augmented matrix of the system until we obtain a matrix in reduced row echelon form.

$$\left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 1 \\ 2 & -1 & 0 & 1 & -2 \\ 1 & -1 & -1 & -2 & 4 \\ 2 & -1 & 2 & -1 & 0 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 1 \\ 0 & -3 & -4 & 1 & -4 \\ 1 & -1 & -1 & -2 & 4 \\ 2 & -1 & 2 & -1 & 0 \end{array}\right] \xrightarrow{R_3 - R_1} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 1 \\ 0 & -3 & -4 & 1 & -4 \\ 0 & -2 & -3 & -2 & 3 \\ 2 & -1 & 2 & -1 & 0 \end{array}\right]$$

$$\xrightarrow{R_4 - 2R_1} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 1 \\ 0 & -3 & -4 & 1 & -4 \\ 0 & -2 & -3 & -2 & 3 \\ 0 & -3 & -2 & -1 & -2 \end{array}\right] \xrightarrow{R_4 - R_2} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 1 \\ 0 & -3 & -4 & 1 & -4 \\ 0 & -2 & -3 & -2 & 3 \\ 0 & 0 & 2 & -2 & 2 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 1 \\ 0 & -2 & -3 & -2 & 3 \\ 0 & -3 & -4 & 1 & -4 \\ 0 & 0 & 2 & -2 & 2 \end{array}\right] \xrightarrow{-\frac{1}{2}R_2} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 1 \\ 0 & 1 & 3/2 & 1 & -3/2 \\ 0 & -3 & -4 & 1 & -4 \\ 0 & 0 & 2 & -2 & 2 \end{array}\right]$$

$$\xrightarrow{R_3+3R_2} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 1 \\ 0 & 1 & 3/2 & 1 & -3/2 \\ 0 & 0 & 1/2 & 4 & -17/2 \\ 0 & 0 & 2 & -2 & 2 \end{array}\right] \xrightarrow{2R_3} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 1 \\ 0 & 1 & 3/2 & 1 & -3/2 \\ 0 & 0 & 1 & 8 & -17 \\ 0 & 0 & 2 & -2 & 2 \end{array}\right]$$

$$\xrightarrow{R_4-2R_3} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 1 \\ 0 & 1 & 3/2 & 1 & -3/2 \\ 0 & 0 & 1 & 8 & -17 \\ 0 & 0 & 0 & -18 & 36 \end{array}\right] \xrightarrow{-\frac{1}{18}R_4} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 1 \\ 0 & 1 & 3/2 & 1 & -3/2 \\ 0 & 0 & 1 & 8 & -17 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right]$$

$$\xrightarrow{R_3-8R_4}\left[\begin{array}{cccc|c}1&1&2&0&1\\0&1&3/2&1&-3/2\\0&0&1&0&-1\\0&0&0&1&-2\end{array}\right]\xrightarrow{R_2-R_4}\left[\begin{array}{cccc|c}1&1&2&0&1\\0&1&3/2&0&1/2\\0&0&1&0&-1\\0&0&0&1&-2\end{array}\right]$$

$$\xrightarrow{R_2-\frac{3}{2}R_3}\left[\begin{array}{cccc|c}1&1&2&0&1\\0&1&0&0&2\\0&0&1&0&-1\\0&0&0&1&-2\end{array}\right]\xrightarrow{R_1-2R_3,\ R_1-R_2}\left[\begin{array}{cccc|c}1&0&0&0&1\\0&1&0&0&2\\0&0&1&0&-1\\0&0&0&1&-2\end{array}\right]$$

From this final matrix, we can read the solution of the system. It is

$$\boxed{A=1,\quad B=2,\quad C=-1,\quad D=-2.}$$